Write Equations to Represent Relationships Between Two Variables

Write an equation for a situation with a rate that multiplies and a start amount that adds.

Name _____ **Date** _____

Tonight's spark

A parking garage charges $5 to drive in, plus $2 for each hour. A driver parks for 3 hours. How much does the driver pay?

Guess first: is it more than $10, or less?

Hours (x)	Total cost (y)
0	$5
1	$7
2	$9
3	?

+$2

Our class model: rate multiplies, start adds

Try it

Work it out on paper. Write one answer on each line.

1 **A taxi charges $2 for each mile. Which equation gives the cost c for m miles?**

A. c = m + 2 **B.** m = 2c **C.** c = 2 – m **D.** c = 2m

My answer _____

2 **Which equation matches '$5 to start plus $2 per hour' with total y for x hours?**

A. y = 7x **B.** y = 2x + 5 **C.** y = 5x + 2 **D.** y = x + 7

My answer _____

3 **In y = 2x + 5, what is y when x = 3?**

A. 30 **B.** 10 **C.** 11 **D.** 16

My answer _____

4 **In y = 2x + 5, which part is the starting amount?**

A. 5 **B.** 2 **C.** x **D.** y

My answer _____

5 **A club costs $16 per hour, so c = 16h. How much does it cost for 3 hours?**

A. $19 **B.** $48 **C.** $13 **D.** $163

My answer _____

Talk about it

Which part of your equation multiplies and which part is added once? How do you know?

Home mission

Pretend a parking spot costs $5 to enter plus $2 for each hour. Ask your student for the cost of 4 hours. Which number is paid once, and which number repeats?

If we get stuck, ask

- Which amount happens every time, and which happens once?
- Does the rate multiply the variable, or get added to it?
- Test your equation on one row you already know.

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/9-3-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it